Zeyu Gan 甘泽宇



Ph.D. Candidate in Artificial Intelligence

Gaoling School of Artificial Intelligence, Renmin University of China

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Beijing, China | Born June 1999

RESEARCH INTERESTS

Mechanism analysis and theory of Large Language Models (LLMs) — including chain-of-thought reasoning, slow-thinking and policy optimization, synthetic data, LLM alignment, and the theoretical foundations of machine learning.

EDUCATION

Ph.D. in Artificial Intelligence — Sep 2024 – Jun 2028 (expected)

Gaoling School of Artificial Intelligence, Renmin University of China · Advisor: Prof. Yong Liu

M.S. in Computer Application Technology — Sep 2021 – Jun 2024

School of Information, Renmin University of China · Advisor: Prof. Suyun Zhao

B.S. in Computer Science — Sep 2017 – Jun 2021

School of Information, Renmin University of China

PUBLICATIONS

[1] **Zeyu Gan**, Zixuan Gong, Yong Liu. Harness Evolution as Learning: Approximation, Generalization, and Optimization Limits of Self-Improving Personal Agents. *arXiv:2609.36892*, 2026.

[2] Zixuan Gong, **Zeyu Gan**, Jiaye Teng, Yong Liu. Normalize-Then-Precondition: A Hierarchical Approach to Marginal Scale and Interaction Geometry for LLM Training. *arXiv:2609.36692*, 2026.

[3] **Zeyu Gan**, Huayi Tang, Yong Liu. Statistical Priors for Implicit Preferences: Decoupling Skill Selection as a Local Harness in Personal Agents. *Findings of the Association for Computational Linguistics: EMNLP*, 2026. EMNLP 2026 (Findings)

[4] **Zeyu Gan**, Ruifeng Ren, Wei Yao, Xiaolin Hu, Gengze Xu, Chen Qian, Huayi Tang, Zixuan Gong, Xinhao Yao, Pengwei Tang, Zhenxing Dou, Yong Liu. Beyond the Black Box: A Survey on the Theory and Mechanism of LLMs. *arXiv:2601.02907*, 2026.

[5] Guoqing Wang, Sunhao Dai, Guangze Ye, **Zeyu Gan**, Wei Yao, Yong Deng, Xiaofeng Wu, Zhenzhe Ying. Information Gain-based Policy Optimization: A Simple and Effective Approach for Multi-Turn LLM Agents. *International Conference on Learning Representations (ICLR)*, 2026. ICLR 2026

[6] **Zeyu Gan**, Hao Yi, Yong Liu. Why Does Reasoning Length Converge? Unveiling the Underfitting-Overfitting Trade-off in Chain-of-Thought. *arXiv:2509.04027*, 2025.

[7] **Zeyu Gan**, Yun Liao, Yong Liu. Rethinking External Slow-Thinking: From Snowball Errors to Probability of Correct Reasoning. *International Conference on Machine Learning (ICML)*, 2025. ICML 2025

[8] **Zeyu Gan**, Yong Liu. Towards a Theoretical Understanding of Synthetic Data in LLM Post-Training: A Reverse-Bottleneck Perspective. *International Conference on Learning Representations (ICLR)*, 2025. ICLR 2025

[9] Pan Du, Suyun Zhao, Puhui Tan, Zisen Sheng, **Zeyu Gan**, Hong Chen, Cuiping Li. Towards a Theoretical Understanding of Semi-Supervised Learning Under Class Distribution Mismatch. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2025. TPAMI

[10] **Zeyu Gan**, Suyun Zhao, Jinlong Kang, Liyuan Shang, Hong Chen, Cuiping Li. Superclass Learning with Representation Enhancement. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. CVPR 2023

HONORS & AWARDS

- Outstanding Master's Thesis (top 10%), Renmin University of China — 2024
- Outstanding Graduate (top 10%), Renmin University of China — 2024
- Outstanding Postgraduate Scholarship (First Class, top 10%) — 2023
- College Student Innovation Experiment Program (National Project) — 2019

INDUSTRY EXPERIENCE

Research Intern, Machine Learning Platform Dept. (TEG), Tencent Aug 2023 – Nov 2023

Worked on WeChat Moments advertising recommendation algorithms; studied large-batch deployment and optimizer design.

ACADEMIC SERVICE

Conference Reviewer

ICLR ICML CVPR AAAI

Award

Gold Reviewer of ICML 2026